

LOCALIZATION IN THE COMMUTANT, STAGE 15:

deformed products, quasi-free lifts and the error cocycle.

The estimate as a renormalized character, the network as a random circuit,
and what a theory-native estimator must compute

10 October 2026

Abstract

This stage starts from the problem, not from the chain. It takes two papers at their core and asks what they say about computing $\mathbb{E}_x f_W(x)$ for one sampled network W .

- Ebrahimi-Fard, Patras, Tapia and Zambotti (arXiv:1710.00735): a law is a deformation of the product, Wick polynomials are the action $\mu^{-1} \star \text{id}$, and renormalization is centering by a character of the noise.
- Zhou (arXiv:2203.15920), after Kuan: a classical interacting process with a reflecting wall is the commutative (Gelfand–Tsetlin) shadow of a free non-commutative random walk $P_t = (\text{id} \otimes \langle \cdot \rangle_t) \Delta$.

Both run on one mechanism, a dynamics as convolution with a family of functionals through a coproduct. We read the network with it, and with the quantum-simulation literature, where the same questions have sharp answers.

- *Wall jets (proved; checked).* Under the *true* law of a pre-activation, the Wick (Appell) coefficients of RELU are the jets of the density at the wall: the transverse measure of the leaf and its normal derivatives. The joint cumulants of RELU outputs are sums of diagrams with wall-jet vertices and joint-cumulant hyperedges, no hyperedge inside one vertex. On a non-Gaussian pair this expansion is $30\times$ more accurate than the Gaussian closure. Renormalizing the jets without the hyperedges is *worse* than not renormalizing at all.
- *Rows are random gates (proved).* The rows of a Gaussian weight matrix diagonalize Hermite degree exactly (Mehler). This is the classical counterpart of Pauli-path orthogonality under random gates, and the radial (dilation) mode is its only obstruction.
- *The error cocycle (proved to leading order; measured on official network 0).* Every estimator’s error splits into a coherent modular part and an incoherent sum over layers of row-births transported by the first jet $\Phi(r)$. The sum rule $\text{MSE} \approx \sum_l \text{birth}_l \times \text{gain}_l$ closes to 4% for the Gaussian closure once its collective mode is removed (that mode is 61% of its error), and to 10% for the exact first-order chain, where the collective mode is already gone. 79–95% of the final error is inherited mean error. 81% of the exact chain’s error is born in layers 11–16.
- *No free lunch for local corrections (proved).* Given the computed state, a birth’s anisotropic part has zero conditional mean. Only its isotropic part is predictable from per-neuron features. This explains stage 9’s failed feature regression and bounds what counterterms can do.
- *The quasi-free lift.* The fold $|z|$ is the reflecting wall (type B), and births come only from the fold. The gate gas is the image of the quasi-free Gaussian field under the sign map: arcsine

pairs (Grothendieck–Krivine rounding) plus star cumulants $-(4/\pi^2) \sum_i \prod_{j \neq i} \rho_{ij}$ (checked to 1–7%), the same stars as births.

- *Complexity.* The Gaussian backbone is the Clifford (matchgate) part, folds are magic, and a fold injects $M_\infty = 0.188$ units of non-Gaussian energy per unit of field, so $n M_\infty / \sqrt{2\pi(1+t)}$ per layer. Generic quantum mean estimation would need about $5\times$ the closures’ budget for all outputs, so the resource being exploited is structure.

From these we derive what a theory-native estimator computes: an exact modular sector, an exact linear cocycle, and births computed where birth $\times$ gain is large, decided per instance. Checks: `scripts/verify_stage15.py`.

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1 The picture in one page, and what this stage refuses to assume

1.1 What this stage refuses to assume

Each earlier stage ended with “what this changes for the chain”. That framing assumes the answer: a system that propagates a truncated law from layer to layer, plus corrections suggested by the theory. This stage drops four assumptions.

1. *That an estimator is a law propagator.* We first ask what *any* estimator’s error is made of (Section 3).
2. *That corrections are fitted.* We prove which parts of an error can be predicted from cheap features and which cannot (Corollary 3.3).
3. *That the weights enter only through matrix products.* They are the noise of a regularity structure. Their ensemble expectations are counterterms, and the information about the instance lives in the centred remainder (Section 2.4).
4. *That the system is the same for every instance.* The error cocycle tells, per network, layer and row, where accuracy is born and where it is lost (Section 6).

1.2 One mechanism in two papers, and in the network

EFPTZ (1710.00735)	Zhou, Kuan (2203.15920)	quantum simulation	network
$\phi_\lambda = \lambda \star \text{id} = (\lambda \otimes \text{id})\Delta$	$P_t = (\text{id} \otimes \langle \cdot \rangle_t)\Delta$	a channel	heat / OU semigroup on inputs; the layer map on laws
moment functional μ	the state $\langle \cdot \rangle_t$	a state	the law of z_l
cumulants $\log^* \mu$	$t^{ \pi } \prod \text{Tr}$ (Faà di Bruno)	connected correlators	joint cumulants: the hyperedges
Wick map $\mu^{-1} \star \text{id}$	—	normal ordering	Appell polynomials of the true law; the ReLU's coefficients are wall jets
deformed product $\cdot_\mu$	—	interaction picture	products of neurons under their true law
BPHZ character $\hat{\mu}^{-1}$	—	average over random gates	ensemble expectation over W : the annealed counterterm
centred model	—	—	the quenched, instance-specific part
restriction to a commutative subalgebra	Gelfand–Tsetlin subalgebra	computational basis	the gate (cell) subalgebra of the Gaussian field
reflecting wall (type B/D)	—	—	the fold $ z $, Weyl-chamber quotient
—	—	Pauli weight	jet order (Hermite degree at the wall)
—	—	random-gate orthogonality	Mehler diagonalization by Gaussian rows
—	—	noise damps high weight	higher jets live only near walls
—	—	Clifford/Gaussian part	the Gaussian backbone (μ_l, C_l)
—	—	magic	folds, $M_\infty = 0.188$ per unit field

1.3 The statements

1. **Wall-jet theorem** (Theorem 2.2; proved, checked). Under the true law, the Appell coefficients of $\text{relu}(z - c)$ are $J_1 = P(Z > c)$ and $J_k = (-1)^k p^{(k-2)}(c)$: the density of the transverse measure at the wall and its normal derivatives. Joint cumulants of RELU outputs are diagram sums with jet vertices and cumulant hyperedges.
2. **Consistency** (Corollary 2.3; checked). The jets and the hyperedges belong to one deformation. Keeping one without the other is worse than the Gaussian closure: on the test pair, -2.7×10^{-2} against -1.2×10^{-2} , against $+4 \times 10^{-4}$ with both.
3. **Rows are random gates** (Theorem 3.1; proved). This is the classical form of Pauli-path orthogonality. The dilation is its only obstruction.
4. **The error cocycle** (Theorem 3.2; leading order, measured). Error = coherent modular part + incoherent sum of row-births transported by $\Phi(r)$. The measured sum rules close to 4% and 10%.

5. **No free lunch** (Corollary 3.3; proved). Only isotropic births are predictable from per-neuron features.
6. **Quasi-free lift** (Section 4). The fold is the reflecting wall; births come only from it; the gate gas is arcsine pairs plus stars (checked to 1–7%).
7. **Complexity** (Section 5). The backbone is Clifford-like and the folds are magic, at density $M_\infty/\sqrt{2\pi(1+t)}$ per neuron (checked to 1–9%). Quantum mean estimation is about $5\times$ worse than the closures for all outputs.

1.4 The measured anatomy of an error (official network 0, truth from 10^9 samples)

	Gaussian closure	exact first-order chain (+ radial κ_4)
final MSE	4.06×10^{-6}	3.53×10^{-8}
share inherited from the previous layer's means (first jet)	0.953	0.790
collective (dilation) share of the inherited error, layers 3–16	0.48–0.68	0.00–0.03
cocycle sum rule $\sum_l \text{birth}_l \text{gain}_l$ vs MSE	2.59 vs $4.06 (\times 10^{-6})$	3.20 vs $3.53 (\times 10^{-8})$
same, collective mode removed	1.50 vs $1.57 (\times 10^{-6})$	3.22 vs $3.52 (\times 10^{-8})$
per-layer transport factor of mean error	0.65–1.04	0.65–0.94
share of the MSE born in layers 11–16	0.51	0.81

Three readings.

- The organisers' radial κ_4 and the κ_3 memory remove exactly the coherent part.
- What remains is a random walk of row-births across layers, dominated by the late layers.
- No global correction can touch it; only better births can.

2 Deformed products: the law as a character, the ReLU's Wick coefficients as wall jets

2.1 The essence of Ebrahimi-Fard–Patras–Tapia–Zambotti

Let $H = \mathbb{R}[x_a : a \in A]$ with the coproduct that makes every x_a primitive, $\Delta x_a = x_a \otimes 1 + 1 \otimes x_a$. The dual product is $(\alpha \star \beta)(P) = (\alpha \otimes \beta)\Delta P$, and the unital functionals $G(H) = \{\lambda : \lambda(1) = 1\}$ form a group under $\star$. The paper's points, in the form we use:

1. *Moments and cumulants are a group element and its logarithm:* $\mu = \exp^\star(\kappa)$ is the Leonov–Shiryaev relation, and the cumulants form an infinitesimal character.
2. *A law is a deformation of the product.* The Wick map $W = \mu^{-1} \star \text{id}$ is the unique map with $W(1) = 1$, $\partial_a W = W \partial_a$ and $\mu \circ W = \varepsilon$ on non-constants. It is a Hopf isomorphism onto $(H, \cdot_\mu, \Delta_\mu, \varepsilon_\mu = \mu)$, with $:A_1 A_2: = :A_1: \cdot_\mu :A_2:$. In the deformed algebra the expectation is the counit.

3. *Renormalization is centring by a character of the noise.* On a comodule of trees over a Hopf algebra of forests, the BPHZ character $\hat{\mu}^{-1} = \hat{\mu} \circ \hat{S}$ is the unique one making every non-empty tree centred (their Theorem 9.1). A good deformation must preserve the remaining structure.

The Gaussian input is one such group element. The heat semigroup is $\gamma_t = \exp^*(tq)$ with $q(x_a x_b) = \delta_{ab}$, so $e^{t\Delta/2} = \phi_{\gamma_t}$ and Hermite polynomials are $\gamma^{-1} \star \text{id}$ of monomials. Stage 9's Eldan path and merge constants live in this one-parameter subgroup.

2.2 A layer acts on the character group

Proposition 2.1 (proved). *Let $\mu_l \in G(H_l)$ be the moment functional of z_l .*

1. *A linear map $z' = Wz$ induces $W^* : x'_i \mapsto \sum_j W_{ij} x_j$, which is a morphism of Hopf algebras. Hence $\mu' = \mu \circ W^*$ and $\kappa' = \kappa \circ W^*$: cumulants transform tensorially and no new ones are created.*
2. *A coordinatewise map g induces an algebra map that is not a coalgebra map. It creates cumulants, by the diagram law of Theorem 2.2.*
3. *The Gaussian closure is the projection $\mu \mapsto \exp^*(\kappa_1 + \kappa_2)$ after every layer, a truncation in the Lie algebra. Its neglected part is the vacuum $\nu = \exp^*(\kappa_{\geq 3})$, with $\mu = \gamma \star \nu$ and $\mathbb{E}_\mu P = \mathbb{E}_\gamma[\phi_\nu P]$: the Edgeworth operator is the action of ν .*

Proof. (1) W^* maps primitives to primitives and is multiplicative, and $\log^*$ commutes with Hopf morphisms. (2) $g^*(x_a) = g(x_a)$ is not primitive. (3) $\star$ is commutative, so $\exp^*$ is a homomorphism from $(H^*, +)$ to $(G, \star)$. $\square$

2.3 The ReLU under its true law: wall jets

Theorem 2.2 (wall jets and the diagram law; proved, checked). *Let Z have all moments and a density p that is smooth near c , and let $A_k = W(Z^k)$ be its Appell (Wick) polynomials. Then $A'_k = kA_{k-1}$, $\mathbb{E}A_k = 0$ for $k \geq 1$, and every polynomial f satisfies $f = \sum_k \mathbb{E}[f^{(k)}(Z)] A_k/k!$ exactly. For $f = \text{relu}(\cdot - c)$ the coefficients are*

$$J_0 = \mathbb{E} \text{relu}(Z - c), \quad J_1 = P(Z > c), \quad J_k = \mathbb{E} \delta^{(k-2)}(Z - c) = (-1)^k p^{(k-2)}(c) \quad (k \geq 2).$$

These are the density of the pre-activation at its wall and its normal derivatives. In the language of stages 11 and 14, they are the jet of the transverse measure of the leaf $\{z = c\}$. For a pair, with J^X, J^Y taken at the respective walls,

$$\begin{aligned} \text{Cov}(f(X), g(Y)) &= \sum_{p, q \geq 1} \frac{\mathbb{E} f^{(p)}(X) \mathbb{E} g^{(q)}(Y)}{p! q!} \mathbb{E}[A_p(X) A_q(Y)], \\ \mathbb{E}[A_p(X) A_q(Y)] &= p! q! [s^p t^q] \exp \left(\sum_{a, b \geq 1} \frac{\kappa_{ab} s^a t^b}{a! b!} \right). \end{aligned}$$

Equivalently, $\mathbb{E}[A_p A_q]$ is the sum over partitions of the $p + q$ legs in which every block meets both vertices, with weight $\prod \kappa(\text{block})$. For m neurons, the joint cumulants of $f_i(X_i)$ are the connected diagrams with jet vertices and joint-cumulant hyperedges, where no hyperedge lies inside one vertex.

Proof. The Appell properties are EFPTZ Theorem 1.1. The expansion of f follows from uniqueness in the Appell basis, by matching $\mathbb{E}f^{(k)}$. For relu , $\text{relu}'' = \delta$. For the pair, $\sum_p s^p \mathbf{A}_p(X)/p! = e^{sX}/\mathbb{E}e^{sX}$, so $\mathbb{E}[e^{sX} :: e^{tY}] = \exp(K(s, t) - K(s, 0) - K(0, t))$, where K is the joint cumulant generating function. Only mixed cumulants survive. Faà di Bruno gives the partitions. $\square$

For a Gaussian pair, only κ_{11} survives and $\mathbb{E}[\mathbf{A}_p \mathbf{A}_q] = \delta_{pq} p! \kappa_{11}^p$ (Mehler). The jets are then $s^{1-k} \text{He}_{k-2}(-\alpha) \varphi(\alpha)$, which is the Gaussian closure's arc-cosine series. Every *chain* is a partial resummation of Theorem 2.2 around Gaussian jets.

Check (Section 7). Take $X = g_1 + 0.15(g_1^2 - 1)$ and $Y = g_2 - 0.1(g_2^2 - 1)$ with $\rho(g_1, g_2) = 0.6$; then $\kappa_{30} = 0.93$, $\kappa_{22} = -0.10$. The diagram law for $\mathbb{E}[\mathbf{A}_p \mathbf{A}_q]$, $p, q \leq 7$, holds to 2×10^{-13} , and $\text{Cov}(X^3 - 2X, Y^4 + Y)$ to ten digits. For $\text{Cov}(\text{relu}(X - 0.3), \text{relu}(Y + 0.2)) = 0.168248$:

order $\leq$	1	2	3	4	5	7
jets + all hyperedges	-4.7×10^{-2}	-8.9×10^{-3}	$+1.6 \times 10^{-3}$	$+8.2 \times 10^{-4}$	$+3.7 \times 10^{-4}$	$+4.0 \times 10^{-4}$
jets + κ_{11} only	-4.7×10^{-2}	-2.6×10^{-2}	-2.7×10^{-2}	-2.7×10^{-2}	-2.7×10^{-2}	-2.7×10^{-2}

The Gaussian closure errs by -1.2×10^{-2} . The full expansion is asymptotic. Its floor of about 4×10^{-4} is set by the nearest singularity of the marginal density, the fold caustic of $g \mapsto g + 0.15(g^2 - 1)$, at distance 2.1 from the wall.

Corollary 2.3 (consistency of a deformation; checked). *The jets and the hyperedges are two faces of one deformation, the Wick map of the true law. Renormalizing one without the other changes the product without changing the counit, and there is no reason for it to help. On the test pair it doubles the Gaussian closure's error (-2.7 against -1.2×10^{-2}), while the consistent expansion reduces it 30-fold. EFPTZ's demand that a deformation preserve the remaining structure is not decoration.*

2.4 Weights as the noise: BPHZ for an estimator

In regularity structures the model Π_ε is built from a noise, the counterterm is the noise expectation, and the renormalized model is the centred one. Our noise is the weights.

- Let $F(W_l; U)$ be any quantity an estimator forms at layer l , polynomial in the fresh matrix W_l given the upstream state U .
- Its character value $\lambda_U(F) = \mathbb{E}_{W_l}[F \mid U]$ is the annealed counterterm.
- $F - \lambda_U(F)$, with nested subtractions given by the twisted antipode, is the centred (quenched) model.

Stages 9 and 13 found that quenched classes win and annealed controls carry nothing. This is the BPHZ split: the annealed part is already inside the Gaussian backbone as its mean field, and the instance's information lives only in the centred parts. Theorem 3.1 below says these centred parts are orthogonal across distinct row multi-indices. That is why their contributions add in quadrature.

3 Rows are random gates: the error cocycle

3.1 Gaussian rows diagonalize Hermite degree

Theorem 3.1 (rows are random gates; proved). *Let W have i.i.d. $N(0, \sigma^2)$ entries, independent of an upstream σ -algebra U . Let a, b be U -measurable vectors with $\rho = \cos \angle(a, b)$, and write $\hat{a} = a/(\sigma|a|)$, $\hat{b} = b/(\sigma|b|)$.*

- *For one row w : $\mathbb{E}[\text{He}_p(w \cdot \hat{a})\text{He}_q(w \cdot \hat{b}) \mid U] = \delta_{pq} p! \rho^p$.*
- *Distinct rows are independent. Hence, for multi-indices $\alpha \neq \beta$ over the output neurons, $\mathbb{E}[\text{He}_\alpha(W\hat{a})\text{He}_\beta(W\hat{b}) \mid U] = 0$.*

If a fixed scale c is used in place of $|a|$, as any basis chosen in advance must do, degree 2 leaks into degree 0 by $\mathbb{E}[\text{He}_2(w \cdot a/\sigma c)] = |a|^2/c^2 - 1$, and every even degree leaks similarly. On the sphere, that is with the homogeneity used, the orthogonality is exact.

Proof. $(w \cdot a, w \cdot b)$ is centred bivariate normal with the stated correlation. Mehler's formula applies. $\square$

This is the classical counterpart of the orthogonality of Pauli paths under random single-qubit gates. That orthogonality is what makes truncated Pauli-path and Pauli-propagation estimates of random circuits accurate on average: their mean squared error is the weight of the dropped paths. Here a Gaussian row plays the random gate. The one thing that couples paths is the norm of the activation vector, which is the dilation (stages 12–14). *Check:* at $\rho = 0.873$, all $p, q \leq 3$ agree within $|z| \leq 1.4$ (4×10^5 rows); the radial leak is 0.2346 predicted against 0.2364 measured.

3.2 The error cocycle

Fix any estimator with layer means $\hat{m}_l$ and fields r_l . Let $\delta_l = \hat{m}_l - m_l$ be its error in $\mathbb{E}h_l$. Define the first-jet transport $T_l \delta = \Phi(r_l) \odot (W_l \delta)$ and the *birth* $\beta_l := \delta_l - T_l \delta_{l-1}$. Then, exactly,

$$\delta_L = \sum_l T_L \cdots T_{l+1} \beta_l.$$

Theorem 3.2 (the cocycle; leading order in the law error). *1. Births live at walls. By Theorem 2.2 applied to $z = w \cdot h_{l-1}$,*

$$\beta_{l,i} = \sum_{k \geq 2} \frac{J_k(r_{l,i})}{k!} \langle \delta \kappa_k(h_{l-1}), w_{l,i}^{\otimes k} \rangle + \dots$$

Here $J_k = s^{1-k} \text{He}_{k-2}(-r) \varphi(r)$ is concentrated near $r = 0$. Only the first jet $\Phi(r)$, which transports, is spread out.

2. Births are i.i.d. over rows given the upstream state, being functions of the fresh row $w_{l,i}$.
3. Births from different layers are orthogonal, except along the dilation. *Two transported births first meet at a fresh row. Their cross term there is $\mathbb{E}[(w \cdot x)\beta(w)] = \sigma^2 x \cdot \mathbb{E} \nabla \beta$ (Stein), which carries no Hermite-degree match and is small. The exception is a dilation $\delta_{l-1} = \varepsilon m_{l-1}$. The first two jets together carry it to $\varepsilon(\Phi \mu + s \varphi) = \varepsilon m_l$ exactly (homogeneity), so its births are coherent from layer to layer.*

4. Hence $\text{MSE} \approx \|\text{coherent modular part}\|^2 + \sum_l \|T_L \cdots T_{l+1} \beta_l^\perp\|^2/n$.

Sketch. (1) Expand $\mathbb{E} \text{relu}(w \cdot h)$ to first order in the difference between the estimated and true laws of h_{l-1} . The first jet is removed into T_l . (2) Rows are i.i.d. given U . (3) Stein’s lemma at the first common fresh layer; homogeneity of relu for the dilation. (4) Expand the square and use (2)–(3). $\square$

The theorem is the classical form of the random-circuit result. Dropping any piece of an estimator costs, additively, the energy of the births it no longer prevents, times their downstream gain. Stage 9’s single-age ladder and its finding that “ages do not cancel” (cumulative norm linear) are instances.

3.3 Measured on official network 0

Truth is from 10^9 samples at every layer; its noise is about 8×10^{-11} at the output.

layer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Gaussian closure: birth $\times$ downstream gain (10^{-7})																
	.00	.28	.75	1.1	1.3	1.5	1.8	2.0	2.1	1.9	2.5	2.4	1.9	2.2	2.3	1.9
exact first-order chain: birth $\times$ downstream gain (10^{-9})																
	.02	.01	.05	.13	.26	.50	.69	1.2	1.4	1.8	2.5	3.5	3.1	3.6	6.0	7.4

- **First jet.** The first jet carries 95% (Gaussian closure) and 79% (exact chain) of the final MSE.
- **Transport factors.** Per layer they are 0.65–1.04 and 0.65–0.94: near-critical.
- **The collective mode.** It holds 48–68% of the Gaussian closure’s inherited error at every layer, against 0–3% for the exact chain.
- **Sum rule.** It misses for the Gaussian closure (2.59 against 4.06×10^{-6}) exactly because of the collective mode. With that mode projected out at every layer it closes to 4% (1.50 against 1.57×10^{-6}). For the exact chain it closes to 10% (3.20 against 3.53×10^{-8}).
- **Where errors are born.** The exact chain’s births grow with depth: 81% of its MSE is born in layers 11–16, and 42% in the last two.

Corollary 3.3 (no free lunch for local corrections; proved). *Given the upstream state and the row’s field r_i , $\mathbb{E}[\beta_{l,i} \mid r_i, U] = \beta^{\text{iso}}(r_i)$. This is a function of the field determined by the isotropic ($O(n)$ -invariant) part of the law error. The anisotropic remainder has conditional mean zero. It is unpredictable from the field, from s_i , or from any per-neuron statistic of the chain, unless the estimator already knows the row-projected law error, in which case it should have corrected the law.*

Proof. Condition on the projection of w_i on the mean direction. The orthogonal part of w_i is a fresh isotropic Gaussian, and it enters only through $\langle \delta \kappa_k, w^{\otimes k} \rangle$, whose conditional mean is a trace. $\square$

At the last layer, after removing the first jet, the part of the error explained by any smooth function of the field is 19% (Gaussian closure) and 7% (exact chain). This settles three earlier findings:

- stage 9's ridge regression on 34 local features failed across networks;
- v56's 103 counterterms are worth -3.8% and no more;
- the isotropic part is the modular sector, which should be *computed* (stages 12–14), not fitted.

3.4 Lattice reading

Stage 14 predicted, by analogy with Rogers' second-moment formula, that the rows are independent given the state, correlated only through the shared collective mode. The cocycle sum rule is the measured form of that statement. It holds once the collective mode is removed and fails before.

4 The quasi-free lift: reflecting walls, folds and the gate gas

4.1 The essence of Zhou's paper (after Kuan)

Zhou studies a random surface growth in the quarter plane: interlacing particles that push and block, with a reflecting wall at 0. It is a one-parameter family of Plancherel measures for $O(\infty)$, so its characters are of type B/D. Three results matter here.

1. *The process is determinantal* along space-like paths, through a generalized L -ensemble.
2. *It is the shadow of a free non-commutative walk.* On $U(\mathfrak{so}_{N+1})$ take the states $\langle X \rangle_\kappa = D(X)\kappa(I)$, with $\kappa_t(O) = e^{t\text{Tr}(O-I)}$. Their moments are an exponential of a trace,

$$\langle F_{i_1 j_1} \cdots F_{i_m j_m} \rangle_t = \sum_{\pi} t^{|\pi|} \prod_B \text{Tr} \prod_{b \in B} F_{i_b j_b},$$

which is Faà di Bruno again. Then $P_t = (\text{id} \otimes \langle \cdot \rangle_{\kappa_t}) \circ \Delta$ is a semigroup that preserves the centre. Through the Harish-Chandra isomorphism, central elements act by symmetric polynomials in the shifted labels, i.e. the particle positions. Restricted to the centre, $P_{t/2}$ is the particle process of one level (Theorem 4.2). Restricted to the Gelfand–Tsetlin subalgebra, generated by the centres of the nested chain $\mathfrak{so}_2 \subset \cdots \subset \mathfrak{so}_{N+1}$, it is the whole multi-level process (Theorem 4.3).

3. *Gaussian free field, and the wall's type matters.* Moments of central elements converge to a three-dimensional Gaussian free field. That field differs from the one of overlapping Wishart spectra, although both live to the right of a wall at 0.

The lessons are these.

- (L1) A process with complicated classical interactions (blocking, pushing, reflection) is the commutative shadow of a free non-commutative process. One computes upstairs.
- (L2) The shadow is taken on a commutative subalgebra built from a nested chain.
- (L3) What kind of wall it is (reflecting, or a hard edge) decides the fluctuations.

The mechanism is the one of Section 2: $P_t = (\text{id} \otimes \langle \cdot \rangle_t) \Delta$ and $\phi_\lambda = (\lambda \otimes \text{id}) \Delta$ coincide for cocommutative Δ . A dynamics is convolution with a semigroup of functionals.

4.2 The network's lift: the quasi-free field and its gate shadow

For the network the upstairs object is the quasi-free (Gaussian) field of pre-activations, the backbone (μ_l, C_l) .

- The commutative subalgebra is generated by the gates, i.e. the cell indicators. This is the diagonal of stage 11's tile groupoid.
- These algebras increase with depth, matching $\text{Gap}_l \subset \text{Gap}_{l+1}$ in stage 14. That is a nested chain as in (L2).
- The gate gas is the restriction of the field's state to it.

Proposition 4.1 (the fold; proved). $\text{relu} = \frac{1}{2}(\text{id} + |\cdot|)$.

1. *The identity half is linear. It commutes with the $O(n)$ symmetry of the weight ensemble, sends quasi-free laws to quasi-free laws, and creates no cumulants (Proposition 2.1).*
2. *The fold $|\cdot|$ is the quotient by the sign group $(\mathbb{Z}/2)^n$. Together with the relabelling symmetry S_n of hidden units, this is the hyperoctahedral group B_n , the Weyl group of type B.*
 - *In the layer's own frame $z = Wx$ the walls are the reflection hyperplanes of $(\mathbb{Z}/2)^n$, and the cells are its chambers.*
 - *So the network alternates $O(n)$ -equivariant linear maps with B_n -equivariant reflecting walls.*
 - *All births come from the folds.*
3. *At a fold, the even and odd Hermite sectors are Laguerre sectors with half-integer parameters:*
 - $\text{He}_{2k}(x) = (-2)^k k! L_k^{(-1/2)}(x^2/2)$ and $\text{He}_{2k+1}(x) = (-2)^k k! x L_k^{(1/2)}(x^2/2)$.
 - *The parameters $a = \pm \frac{1}{2}$, which index Zhou's two level parities through the Jacobi weights $(1-x)^a(1+x)^{-1/2}$, are the spectral signature of a reflecting wall. Our fold carries the same pair in its Laguerre limit.*
4. *The folded law is an image sum, $p_{|z|}(y) = p(y - \mu) + p(y + \mu)$ for $y > 0$. This is the method of images of the reflecting walk.*

4.3 The gate gas: arcsine pairs and stars

Proposition 4.2 (proved to leading order; checked). *Let $z \sim N(0, R)$ have unit variances and correlations ρ_{ij} , and let $\varepsilon_i = \text{sign } z_i$. Then $\mathbb{E}\varepsilon_i \varepsilon_j = \frac{2}{\pi} \arcsin \rho_{ij}$, and the fourth joint cumulant is*

$$\kappa(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) = -\frac{4}{\pi^2} \sum_{i=1}^4 \prod_{j \neq i} \rho_{ij} + O(\rho^4).$$

Proof. Write $\text{sign} = \sum_{k \text{ odd}} a_k \text{He}_k/k!$ with $a_k = 2\text{He}_{k-1}(0)\varphi(0)$, so $a_1 = \sqrt{2/\pi}$ and $a_3 = -\sqrt{2/\pi}$. In the Gaussian diagram expansion the two-vertex components reproduce the arcsine pairings. The lowest connected diagram with odd degrees on four vertices is the star of degrees $(3, 1, 1, 1)$, of weight $a_3 a_1^3 = -(2/\pi)^2$. $\square$

Check (orthant probabilities by Genz quadrature): the ratio of the exact cumulant to the star formula is 1.013, 1.038, 1.074 at correlation scales 0.1, 0.2, 0.3. Two readings follow.

- *Rounding.* The arcsine law is Grothendieck–Krivine rounding of the linear, quasi-free correlation. Tsirelson’s theorem realizes the linear correlations themselves with Clifford observables. The classical gates pay the rounding, up to the Grothendieck constant in correlation strength.
- *Stars are the non-quasi-free content.* What the gate gas has beyond pairings begins with stars, gates with three legs. These are the hub-rooted stars of stage 9 (P2) and the births of the chain. Stage 12 called the memory the “non-quasi-free remainder”; this is its first diagram.

4.4 Two sectors, and where the missing information lives

Every statistic an estimator forms from the instance splits into two parts.

- An $O(n)$ -invariant part: spectral and Gram data, traces, kernels. This is the isotropic sector, and it contains the modular one.
- A B_n -specific remainder, which depends on the neuron basis: Hadamard and diagonal readouts, hub pairings.

For the invariant sector, (L1)–(L2) apply in their usual form. Linear spectral statistics of nested rank-one (hub) chains have Gaussian fluctuations whose covariance is computable from the ensemble, as for Wishart minors. For the B_n sector no such reduction exists. Corollary 3.3 says that this is where the information an estimator lacks lives. Lesson (L3) appears as the split $\text{relu} = \frac{1}{2}(\text{id} + |\cdot|)$. The identity half carries the first jet, so it transports errors; the fold half creates them.

5 Complexity: a random circuit with a Gaussian backbone

5.1 Gaussian backbone, magic folds

A Gaussian law under linear maps is simulated exactly by (μ, C) . This is the classical counterpart of Gottesman–Knill (stabilizer states under Clifford gates) and of fermionic linear optics (match-gates). Folds are the only non-Gaussian gates. Their “magic” at field r is the energy above the first chaos, which is stage 14’s kink law summed over orders:

$$m(r) = \text{Var } \text{relu}(r + \xi) - \Phi(r)^2 = \sum_{k \geq 2} \frac{\text{He}_{k-2}(r)^2 \varphi(r)^2}{k!}, \quad m(0) = \frac{1}{2} - \frac{1}{2\pi} - \frac{1}{4} = 0.0908.$$

Proposition 5.1 (the magic budget; checked). $M_\infty = \int m(r) dr = 0.18806$. The order k contributes $\int E_k = 0.141, 0.024, 0.009, 0.004, \dots$ for $k = 2, 3, 4, 5$. Under stage 12’s field law $r \sim N(0, t_l)$, the magic per neuron is $\mathbb{E} m(r) \simeq M_\infty / \sqrt{2\pi(1 + t_l)}$:

- 0.0577 against 0.0531 at $t = 1$;
- 0.0346 against 0.0336 at $t = 4$;
- 0.0211 against 0.0208 at $t = 12$.

So a late layer of the competition networks injects about $0.021 n \approx 22$ units, and the total over depth grows like $\log \tau$ (stage 12’s harmonic Fatou law).

For circuits dominated by Clifford gates, the cost of exact simulation is exponential in the magic. The mean, however, needs only connected diagrams of low order. The cumulant (birth) expansion is a low-magic expansion. Its cost is set by how many magic insertions must be tracked *coherently*, which for a chain is its number of live κ_3 families. Theorem 3.2 says that incoherent magic costs only its energy.

5.2 The Pauli-propagation dictionary

random quantum circuit	bias-free He RELU network
Pauli weight of a path	jet order k (Hermite degree at a wall)
random single-qubit gate	a fresh Gaussian row (Theorem 3.1)
paths orthogonal on average	births orthogonal across rows and layers (Theorem 3.2)
noise damps weight k by $(1-p)^k$	jets $k \geq 2$ live at walls: $\mathbb{E}_r[s^{2k-2}J_k^2] \propto (1+t)^{-1/2}$, algebraic, not exponential
weight 1 propagates	the first jet $\Phi(r)$ transports with factor 0.65–0.94, near-critical
truncation MSE = weight of dropped paths	the cocycle sum rule, measured to 4–10%
the one non-orthogonal coupling	the dilation (radial norm)

The network is a weakly damped, near-critical random circuit. Damping is algebraic, through the wall density, so memory persists over many layers (stage 9: eleven ages matter). Births are not damped where they are born, so the late layers dominate.

5.3 Calibration against quantum mean estimation

Take $\sigma^2 = 0.0787$ (average output variance), one forward pass = $2Ln^2$, and $B = 2048n^3$. Quantum mean estimation needs about σ/ε coherent passes per output, and about $\sqrt{n}\sigma/\varepsilon$ for all n outputs. The latter is the multi-observable bound. We ignore reversibility and error-correction overheads, which is optimistic for the quantum side.

target raw MSE	classical Monte Carlo	quantum, one output	quantum, all outputs	deterministic closures
1.56×10^{-8}	77 B	0.034 B	1.10 B	about 0.2 B (v56)
1.0×10^{-9}	1201 B	0.135 B	4.33 B	—

For all outputs, the closures beat generic quantum mean estimation by about $5\times$. They exploit structure that a generic algorithm does not: one covariance gives every output at once, the Gaussian backbone. Large further gains can only come from more structure. They cannot come from better generic sampling, classical or quantum.

5.4 The low-degree view in weight space

Average the MSE over networks. In L^2 of the weight ensemble the truth and any estimator have Hermite (weight-chaos) expansions, and errors of different weight-degree are orthogonal. Corollary 3.3 says the following in this language.

- Corrections built from features of degree 0 in the anisotropic directions of a fresh row reach only the isotropic births.

- An anisotropic birth at row i is a polynomial of degree $k \geq 1$ in those directions, contracted with the instance's $\delta\kappa_k$.
- The hardness of the problem is the number of anisotropic cumulant directions that must be resolved at the late layers. Global calibration cannot change it.

5.5 Coding

- *A layer is a one-bit embedding.* $E[d_H(\text{sign } Wx, \text{sign } Wy)] = n\theta(x, y)/\pi$. Each layer one-bit-senses the previous one. The closure's two-point law is the code's distance law, which is the stage-12 angle map.
- *Two Grams of one operator.* Let $(\mathcal{G}f)_i = \mathbb{E}[\bar{g}_i f]$ with centred gates $\bar{g}_i$.
 - $\mathcal{G}\mathcal{G}^*$ is the neuron-side gate covariance, which the closure computes.
 - $\mathcal{G}^*\mathcal{G}$ is the input-side code kernel $\sum_i \bar{g}_i(x)\bar{g}_i(y)$, which NNGP theory computes.
 - They have the same non-zero spectrum. The neuron picture and the input picture are dual codes of one operator, a MacWilliams-type duality.
- *Lattice side.* The cell masses are the code's weight distribution (stage 14's gap labels). The tight frames of contact normals are its local codes.

6 What a theory-native estimator computes

We now derive the estimator from Sections 2–5, and from nothing else.

6.1 Three sectors with three different statuses

By Theorem 3.2, an estimator's error has three parts, and each needs a different treatment.

S1, the modular sector: compute exactly. The dilation is the one coherent mode. The first two jets carry it with factor exactly 1, and its errors add in amplitude across layers. It was 61% of the Gaussian closure's MSE. It must not be approximated or fitted. Use stage 13's dilation package (exact to $O(\varepsilon^2)$ on scale mixtures), or stage 14's radial partition function, at $O(n^2)$ per layer.

S2, the linear cocycle: apply exactly. $T_l = \Phi(r_l) \odot W_l$ is a matrix–vector product. Its adjoint gives each row's downstream gain $a_{l,i}$, the norm of row i of $T_L \cdots T_{l+1}$.

S3, the births: the only thing approximated.

$$\beta_{l,i} = \sum_k \mathbf{J}_k(r_{l,i}) \langle \delta\kappa_k, w_{l,i}^{\otimes k} \rangle / k!.$$

Compute them with true-law jets *and* the matching hyperedges, at one consistent order (Corollary 2.3), and only where birth $\times$ gain is large.

This is the interaction picture. The Gaussian backbone is the free field. Non-Gaussianity enters only as births, and their effect on later means is linear and exact.

6.2 The instance decides the estimator

The question was how an optimal system could fail to couple the sampled weights to the shape of its own computation. In the cocycle picture the coupling is forced, and it can be computed.

Proposition 6.1 (additivity makes allocation a knapsack). *Under the orthogonality of Theorem 3.2, the expected MSE of an estimator that captures the fraction $c_{l,i}$ of the birth at (l, i) is*

$$\sum_{l,i} (1 - c_{l,i})^2 a_{l,i}^2 \mathbb{E} \beta_{l,i}^2, \quad \mathbb{E} \beta_{l,i}^2 \approx \sum_{k \geq 2} \frac{(s^{k-1} J_k(r_{l,i}))^2}{k!} \epsilon_{l,k}^2,$$

where $\epsilon_{l,k}$ is the relative row-projected error of the order- k cumulant at layer l . The terms are additive. So the optimal allocation of a FLOP budget over (layer, row, order) is a fractional knapsack, solved greedily by captured energy per FLOP.

The weights come from a cheap first pass that depends on the instance: the fields $r_{l,i}$ and scales $s_{l,i}$ from the Gaussian backbone, and the gains $a_{l,i}$ from one adjoint sweep. Both are $O(n^3)$ once. The second pass spends the budget where the weights say. Different networks then get different estimators, as they should: a network with more near-wall rows at depth gets more late-layer effort. Dynamically, the allocation of layer $l + 1$ can be revised once layer l 's births are known.

6.3 Where the budget goes on the official networks

On network 0 the exact first-order chain's births are 5% in layers 1–6 and 81% in 11–16.

- **Last-layer channels.** A fit of the last layer's residual-variance profile by squared jets, weighted over the actual fields, attributes 59% of the last-layer birth to the second jet (a covariance error of h_{15} , profile φ^2). The third jet (κ_3 , profile $r^2\varphi^2$) takes 31% and the fourth 11%. The fit is coarse: 20 bins, residual 0.30.
- **What causes them.** By Theorem 2.2, covariance births at a late layer come from the $(2, 2)$ and $(3, 1)$ hyperedges and the κ_3 -pair diagrams in $\text{Cov}(\text{relu}, \text{relu})$. These are what the exact chain lacks and what v56's κ_4 sector partly adds. The theory points the effort at late covariance births.

Heisenberg reads for the late layers. The late readouts dominate, so read the old κ_3 families backwards.

- *The read.* Transport the readout rows of a late layer L' to the layer l_f where a family was born, giving $V = (B_{L'-1} \cdots B_{l_f+1})^\top w$. Then evaluate $\sum_a c_a (Vu_a)(Vv_a)^2$ exactly. Old sources need no compression, and stage 9 showed that Frobenius compression fails against Hadamard readouts.
- *Restricting rows.* Only near-wall rows need this ($J_3 \propto r\varphi(r)$): about half at $t \approx 12$.
- *Cost.* About 6 late readout layers, each with backward transport and family reads over at most 16 layers on half the rows, cost on the order of $100 n^3 \approx 0.05 B$.
- *Early layers* keep a short forward window. Their births are small and discounted by the gain.

This is a meet-in-the-middle estimator: Schrödinger (forward) where births are cheap and discounted, Heisenberg (adjoint) where they dominate.

6.4 What this system is not

standard (public chain, v56)	theory-native
one closure, the same at every layer and for every network	three sectors; allocation decided per instance (Proposition 6.1)
law propagator with feedback of corrections into (μ, C)	interaction picture: exact backbone, births transported by the exact cocycle
dilation absorbed in fitted pair classes and counterterms	modular sector computed exactly; counterterms proved useless beyond the isotropic part
old sources compressed forward	late readouts done backward (Heisenberg), exactly, on near-wall rows
Gaussian jets with Edgeworth patches	true-law jets and hyperedges at one consistent order

6.5 Predictions and the next measurements

1. **The modular sector alone.** Making it exact in the Gaussian closure removes at most its collective share, 61% of the MSE. The MSE then goes from 4.1×10^{-6} towards 1.6×10^{-6} , at $O(n^2)$ cost.
2. **Counterterms.** On top of an exact first-order chain, counterterms of any kind gain at most about $0.21 \times 0.07 + 0.01 \approx 2.5\%$.
3. **The last layer.** The true covariance of h_{15} will show that about 60% of the exact chain's last-layer birth is row-projected covariance error. Testing this needs one Monte Carlo run of $\text{Cov}(h_{15})$, about 2×10^6 samples.
4. **Early precision.** Readout and birth precision in layers 1–6 can be relaxed about $3\times$ in amplitude at a cost of $\lesssim 5\%$ of the MSE, provided the sources born there are kept for the late Heisenberg reads.
5. **Biased networks** (stage 14's conjecture). With biases the dilation is no longer free. The collective share and the coherent part of the cocycle should then shrink with the bias scale.

7 Checks

The script `scripts/verify_stage15.py` runs in under a minute. Its transcript is `notes/stage15/verify_stage15.txt`. Section 2 needs the official network 0 and its 10^9 -sample truth. The exact first-order chain is the organisers' $K = 3$ simple chain with radial κ_4 , ported and verified in stage 9 (MSE 3.53×10^{-8}).

statement	result
connected-cumulant law $\mathbb{E}[\mathbf{A}_p \mathbf{A}_q]$, $p, q \leq 7$	max relative difference 2.3×10^{-13}
Appell expansion of $\text{Cov}(X^3 - 2X, Y^4 + Y)$	2.3159134053 both ways
wall-jet expansion of $\text{Cov}(\text{relu}, \text{relu})$	errors $-4.7 \times 10^{-2} \rightarrow +4.0 \times 10^{-4}$ (orders $1 \rightarrow 7$), Gaussian closure -1.2×10^{-2}
consistency (jets without hyperedges)	-2.7×10^{-2} , worse than the Gaussian closure
row Mehler, $p, q \leq 3$	$ z \leq 1.4$ over 4×10^5 rows; radial leak 0.2346 vs 0.2364
first-jet share of the final MSE	0.953 (Gaussian closure), 0.790 (exact chain)
cocycle sum rule	2.59/4.06 (Gaussian), 1.50/1.57 after removing the collective mode, 3.20/3.53 (exact chain)
collective share of inherited error	0.48–0.68 (Gaussian), 0.00–0.03 (exact chain)
births in layers 11–16	51% (Gaussian), 81% (exact chain)
last-layer channels (exact chain)	covariance 0.59, κ_3 0.31, κ_4 0.11; field-explained part of the remainder 0.073
gate-gas star cumulant	exact/star = 1.013, 1.038, 1.074 at scales 0.1, 0.2, 0.3
magic M_∞ and the layer law	0.18806; $\mathbb{E}m$ vs $M_\infty/\sqrt{2\pi(1+t)}$: 0.0577/0.0531, 0.0346/0.0336, 0.0211/0.0208
quantum calibration	Monte Carlo 77 B ; quantum 0.034 B (one output), 1.10 B (all outputs) at MSE 1.56×10^{-8}

What is not checked here:

- the Stein cross-term bound in Theorem 3.2, beyond the measured sum rules;
- the knapsack allocation and the Heisenberg reads, which are designs, not results;
- the Laguerre parallel with Zhou’s parities, which is structural, not quantitative.